

Humanness and word order: Evidence from Czech left branch extraction*

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Daniela Kořánová
Charles University
daniela.koranova@ff.cuni.cz

Radek Šimík
Charles University
radek.simik@ff.cuni.cz

Abstract We deliver an experimental argument from Czech, a Slavic language, that humanness-related word order manipulations are of narrowly syntactic origin. In particular, we show that while left branch extraction from objects across subjects is generally allowed, it becomes unavailable when the object denotes a human and the subject a nonhuman. We argue that this follows from an independently proposed account of subject–object asymmetry in subextraction (Ko 2007; 2014), conjoined with the assumptions that (i) human objects must scramble across nonhuman subjects (unless a stronger factor overrides this requirement) and (ii) that the scrambling is movement-based rather than base-generated. At a more general level, our experiments provide novel empirical support for specific syntactic theories (phase-based syntax, cyclic linearization) and contribute to our understanding of the relation between semantic features such as humanness or animacy and core syntactic processes.

Keywords: humanness; word order; scrambling; syntax; phases; linearization; left branch extraction; Czech

1 Introduction

We provide a new, experimentally supported observation that the availability of left branch extraction in Czech is modulated by the humanness of the arguments involved: in clauses where the subject and object are imbalanced in humanness, left branch extraction is only possible from a nonhuman object across a human subject; (1a). Extraction from a human object across a nonhuman subject (1b), or extraction from a subject across an object (1c)/(1d) – irrespective of their humanness – is less acceptable and, as we argue, ungrammatical.¹

- (1) Czech (from our experimental stimuli; judgments reflect experimental results)
Detektiv na začátku vyšetřování tušil, ...
detective at beginning investigation had.idea
‘At the beginning of the investigation, the detective had an idea...’

* Acknowledgments: To be added.

¹ The (linearly discontinuous) constituent affected by the extraction is italicized; the intervener is set in gray. The case information in the glosses is color-coded for better orientation in the pattern: NOM = subject; ACC = object.

- | | | |
|----|--|---------------------|
| a. | <i>kterou</i> <i>kuchařka</i> <i>otrávila</i> <i>omáčku</i> .
which.ACC cook.NOM poisoned sauce.ACC
‘which sauce a/the cook poisoned.’ | WH + SVO / HUM-NHUM |
| b. | * <i>kterou</i> <i>omáčka</i> <i>otrávila</i> <i>kuchařku</i> .
which.ACC sauce.NOM poisoned cook.ACC
Intended: ‘which cook a/the sauce poisoned.’ | WH + SVO / NHUM-HUM |
| c. | * <i>která</i> <i>kuchařku</i> <i>otrávila</i> <i>omáčka</i> .
which.NOM cook.ACC poisoned sauce.NOM
Intended: ‘which sauce poisoned a/the cook.’ | WH + OVS / HUM-NHUM |
| d. | * <i>která</i> <i>omáčku</i> <i>otrávila</i> <i>kuchařka</i> .
which.NOM sauce.ACC poisoned cook.NOM
Intended: ‘which cook poisoned a/the sauce.’ | WH + OVS / NHUM-HUM |

The observation sheds new light on the syntactic relevance and representation of humanness: its non-trivial interaction with a syntactic process like left branch extraction points to the need of representing humanness in narrow syntax, also in cases where it remains morphologically unrealized (by agreement or case-marking). As we will show, the observed pattern also supports a particular view of A-scrambling, namely A-scrambling as movement (Bailyn 2004; Biskup 2011; Antonyuk 2015; a.o.) rather than base-generation (Titov 2012; Neeleman & van de Koot 2002; a.o.).

More specifically, we argue that the observed pattern follows from the conjunction of the following independently motivated assumptions: (i) syntactic derivation proceeds by phase (vP and CP; Chomsky 2001), (ii) linearization proceeds by phase (so-called cyclic linearization; Fox & Pesetsky 2005), (iii) movement from SpecXP to SpecXP is banned (Ko 2007), (iv) human objects/patients scramble over nonhuman subjects/agents (unless the scrambling is avoided for independent reasons; Titov 2012), and (v) base-generation respects the Uniformity of Theta-Assignment Hypothesis (UTAH; Baker 1988), i.e. the human object scrambling across a nonhuman subject is achieved by movement rather than base-generation (see, e.g., Bailyn 2004; Biskup 2011; Antonyuk 2015; cf. Titov 2012). Assumptions (i)–(iii) are adopted as a “package” from Ko (2007; 2014) (see also Bondarenko & Davis 2021) and account for the subject–object asymmetry in left branch extraction, illustrated by the acceptability contrast between (1a) and (1d). The additional assumption (iv) explains the contrast between (1a) and (1b), which would not be expected if humanness played no role, and, finally, assumption (v) explains the contrast between (1a) and (1c), which would not be expected if the arguments could “swap” their positions already in the base-generated structure (as proposed by Titov 2012).

The article is organized as follows. Section 2 discusses some empirical preliminaries. In §2.1 we introduce the phenomenon of left branch extraction and show experimentally that the ban on extraction from subjects across objects (illustrated above by (1c) and (1d)), which has been documented in languages like Japanese, Korean, or Russian and which was one of the empirical motivations for Fox & Pesetsky’s (2005) theory of cyclic linearization, is indeed active in Czech, too. In §2.2 we discuss the effect of humanness on word order and grammar more generally. Section 3 states the core research questions and competing hypotheses concerning the relation between argument humanness and left branch extraction. We demonstrate the differing predictions of the hypotheses, relying on the baseline analysis of Ko (2007; 2014). Section 4 contains the main empirical contribution of this article. It discusses four naturalness rating experiments, shedding light on the questions and hypotheses considered. Section 5 concludes the paper.

2 Basic concepts and observations

2.1 Left branch extraction and the subject–object asymmetry

Left branch extraction (LBE) is a special case of subextraction from nominal phrases, where the extracted element is the “left branch” of the NP, typically an adjective or determiner (Corver 1990; 2017; Fanselow & Ćavar 2002; Bošković 2005; Abels 2012; Murphy 2021; Bondarenko & Davis 2021; 2023; a.o.). There are three main types of analysis of LBE in the literature: based on subextraction, on distributed deletion, and on remnant movement. These analyses are illustrated by the three structures representing example (2), a case of an extraction of an interrogative *wh*-determiner from an object NP.² In the subextraction analysis (e.g., Corver 1990; Bošković 2005; Bondarenko & Davis 2021), the “left branch” *kteřou* ‘which’ moves by itself – via the *vP* phase edge, landing in SpecCP – and strands the rest of the NP, containing the NP *zadání* ‘assignment’, in the VP. In the distributed (aka scattered) deletion analysis (Fanselow & Ćavar 2002; Bondarenko & Davis 2023), crucially relying on the copy theory of movement (Chomsky 1993), the whole NP *kteřé zadání* ‘which assignment’ moves and the movement is followed by the deletion of ‘which’ in VP, ‘assignment’ in SpecCP, and – assuming *vP* is a phase – the whole NP in SpecvP. Finally, the remnant movement analysis (e.g., Bašić 2009; Abels 2012; Veselovská 2018) assumes that the NP ‘assignment’ first moves out the larger NP ‘which assignment’ (we assume it moves to the edge of VP) and then the remnant NP – containing ‘which’ and the trace after the movement of ‘assignment’ – moves to SpecCP (via SpecvP).

- (2) *Czech*
 Které jsi měla zadání?
 which.ACC AUX.2SG had.PTCP assignment.ACC
 ‘Which assignment did you have?’
- Subextraction analysis*
 $[_{CP} \text{ which}_1 [_{TP} \text{ AUX } [_{PTCP} \text{ had } [_{vP} t'_1 t_v [_{VP} t_v [_{NP} t_1 \text{ assignment}]]]]]]]$
 - Distributed deletion analysis*
 $[_{CP} [_{NP} \text{ which assignment}] [_{TP} \text{ AUX } [_{PTCP} \text{ had } [_{vP} [_{NP} \text{ which assignment}] t_v [_{VP} t_v [_{NP} [_{NP} \text{ which assignment}]]]]]]]$
 - Remnant movement analysis*
 $[_{CP} [_{NP} \text{ which } t_1]_2 [_{TP} \text{ AUX } [_{PTCP} \text{ had } [_{vP} t'_2 t_v [_{VP} \text{ assignment}_1 t_v t_2]]]]]$

We believe that our analysis is in principle compatible with all the analytical options, although we do not show it due to lack of space. We will henceforth adopt the subextraction analysis (following Bondarenko & Davis 2021), mainly for the sake of simplicity.

Let us first consider the subject–object asymmetry in subextraction, as discussed and analyzed in Bondarenko & Davis (2021) (who in turn follow Ko 2007; 2014). The contrast in (3) is reminiscent of the contrast between (1a) and (1d): extraction from an object across a subject is acceptable, but extraction from a subject across a (scrambled) object is not.

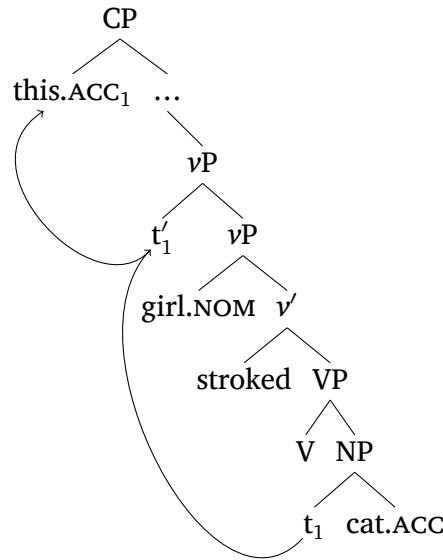
- (3) *Russian* (Bondarenko & Davis 2021: 1)

² Extracting other types of elements like adjectives, demonstratives, or possessives is possible in Czech (Veselovská 2018), but interrogative *wh*-extraction is arguably most common and relatively contextually unrestricted. Non-interrogative extractions are usually associated with a contrastive topic–focus structure.

- a. *Ětogo*₁ *devočka* *pogladila* [*t*₁ *kota*].
 this.ACC girl.NOM stroked cat.ACC
 ‘The girl stroked this cat.’
- b. **Ěta*₁ *kota* [*t*₁ *devočka*] *pogladila*.
 this.NOM cat.ACC girl.NOM stroked
 ‘The girl stroked this cat.’

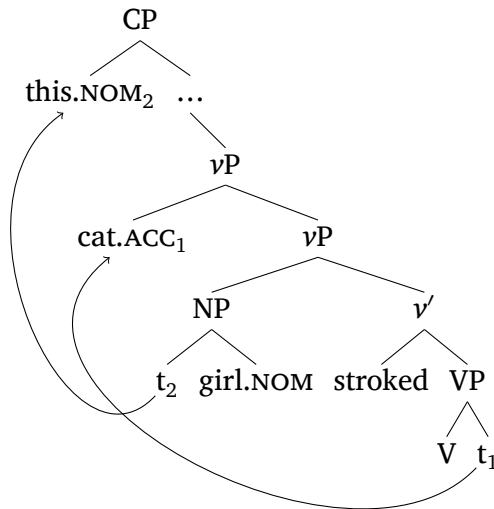
Consider first the derivation of (3a), provided in (4), adopting the assumptions summarized in section 1. On its way to the left periphery (SpecCP), the subextracted demonstrative ‘this’ first moves to the *vP* phase edge (as per Chomsky 2001). When the *vP* is complete, the dominated terminal nodes undergo linearization (as per Fox & Pesetsky 2005), as detailed in (5a). The derivation continues and the demonstrative ‘this’ eventually lands in its final position – SpecCP. There the linearization is repeated, see (5b).

(4) *Extraction from object across subject*



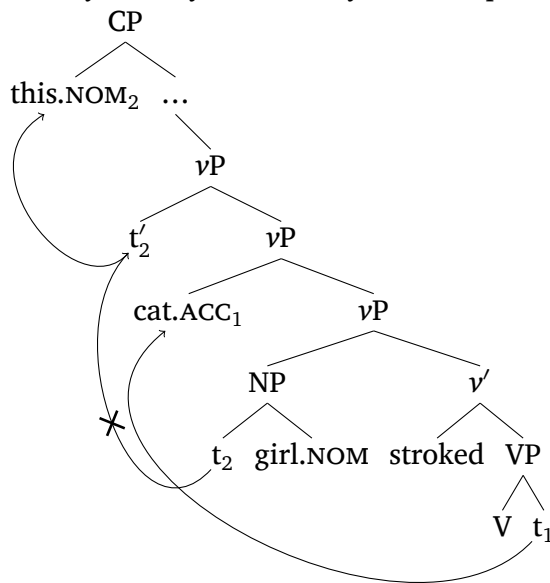
- (5) a. Linearization at *vP*
 (i) this < girl
 (ii) girl < cat
 (iii) this < cat
- b. Linearization at CP
 (i) this < girl
 (ii) girl < cat
 (iii) this < cat

Now consider two ways in which the system correctly fails to derive (3b). The first one is sketched in (6). The object ‘cat’ first scrambles over the subject, landing in Spec*vP*. The linearization procedure at *vP* yields the statements in (7a). After ‘this’ moves to SpecCP, the structure is linearized again, yielding (9b). In result, the linearization statements (7a-i) and (7b-i), framed for clarity, are in contradiction, which corresponds to ungrammaticality in the cyclic linearization theory.

(6) *Extraction from subject across object (attempt 1)*

- (7) a. Linearization at vP
- (i) cat < this
 - (ii) this < girl
 - (iii) cat < girl
- b. Linearization at CP
- (i) this < cat
 - (ii) cat < girl
 - (iii) this < girl

The other failure is shown in (8). The linearization contradiction is lifted by first subextracting ‘this’ from the subject to a higher SpecvP before its final movement to SpecCP (see (9)). Yet, this intermediate movement – call it SPEC-TO-SPEC movement – is banned (as is indicated by the crossed movement path), according to Ko (2007). The rationale behind the ban is that SpecvP is not c-commanded by *v*, which would be required for a movement to SpecvP to take place. Ergo, the subject–object subextraction asymmetry is derived.

(8) *Extraction from subject across object (attempt 2)*

- (9) a. Linearization at vP
- (i) this < cat
 - (ii) this < girl
 - (iii) cat < girl
- b. Linearization at CP
- (i) this < cat
 - (ii) cat < girl
 - (iii) this < girl

In what follows, we provide experimental evidence that Czech, like Russian, but also Japanese (Haig 1980; Saito 1985; a.o.) and Korean (Park & Sohn 1993; Ko 2007; 2014; a.o.), exhibits the same subject–object asymmetry in subextraction. In addition, we also show that the asymmetry only holds if the object is actually overtly realized (rather than dropped), corroborating the linearization-based source of the violation. Finally, we show

that LBE in Czech is only fully acceptable from clause-final constituents (which motivates using clause-final subjects in our main experimental stimuli; cf. Bondarenko & Davis's 2021 Russian example in (3)).³

The above-mentioned contrasts are evident in the results of an experiment whose stimuli were used as fillers for the experiment reported in Šimík et al. (2022). Since the results have not yet been published, we also briefly describe the experimental design.⁴

Design and materials: In their LBE experiment, Šimík and colleagues manipulated three factors (independent variables): EXTRACTION TARGET (subject vs. object), TARGET POSITION (final vs. nonfinal), and INTERVENER (absent vs. present).⁵ They used a fully crossed $2 \times 2 \times 2$ design, with all factors manipulated within items and within subjects and with stimuli distributed to lists using the standard Latin Square design. An example of an item (set of stimuli) in all its 8 conditions is provided in (10). There were 32 items like (10) and they were complemented with an additional 80 items (fillers or items from different experiments).

- (10) Vyptávala se, ...
 asked REFL
 'She asked...'
- | | | |
|----|--|--------------------|
| a. | <i>kterou</i> včera zametal garáž.
which.ACC yesterday swept garage.ACC
'which garage he swept yesterday.' | [obj, fin, abs] |
| b. | <i>kterou</i> soused zametal garáž.
which.ACC neighbor.NOM swept garage.ACC
'which garage a/the neighbor swept.' | [obj, fin, pres] |
| c. | <i>kterou</i> včera garáž zametal.
which.ACC yesterday garage.ACC swept
'which garage he swept yesterday.' | [obj, nfin, abs] |
| d. | <i>kterou</i> soused garáž zametal.
which.ACC neighbor.NOM garage.ACC swept
'which garage a/the neighbor swept.' | [obj, nfin, pres] |
| e. | <i>který</i> včera zametal soused.
which.NOM yesterday swept neighbor.NOM
'which neighbor swept (did the sweeping) yesterday.' | [subj, fin, abs] |
| f. | <i>který</i> garáž zametal soused.
which.NOM garage.ACC swept neighbor.NOM
'which neighbor swept a/the garage.' | [subj, fin, pres] |
| g. | <i>který</i> včera soused zametal.
which.NOM yesterday neighbor.NOM swept
'which neighbor swept (did the sweeping) yesterday.' | [subj, nfin, abs] |
| h. | <i>který</i> garáž soused zametal.
which.NOM garage.ACC neighbor.NOM swept
'which neighbor swept a/the garage.' | [subj, nfin, pres] |

Task, participants, and administration: Participants (96 convenience sampled non-linguists) rated each stimulus on a scale from 1 (completely unnatural) to 7 (completely natural). The experiment was prepared (including Latin Square distribution and pseudo-

³ Starke (2001) observes this for nonfinal (preverbal) subjects, but we show experimentally that the same holds for nonfinal objects.

⁴ We analyze the data kindly provided to us by the authors, with the authors' permission.

⁵ We set variables (factors) in SMALL CAPS and their values (levels) in sans serif.

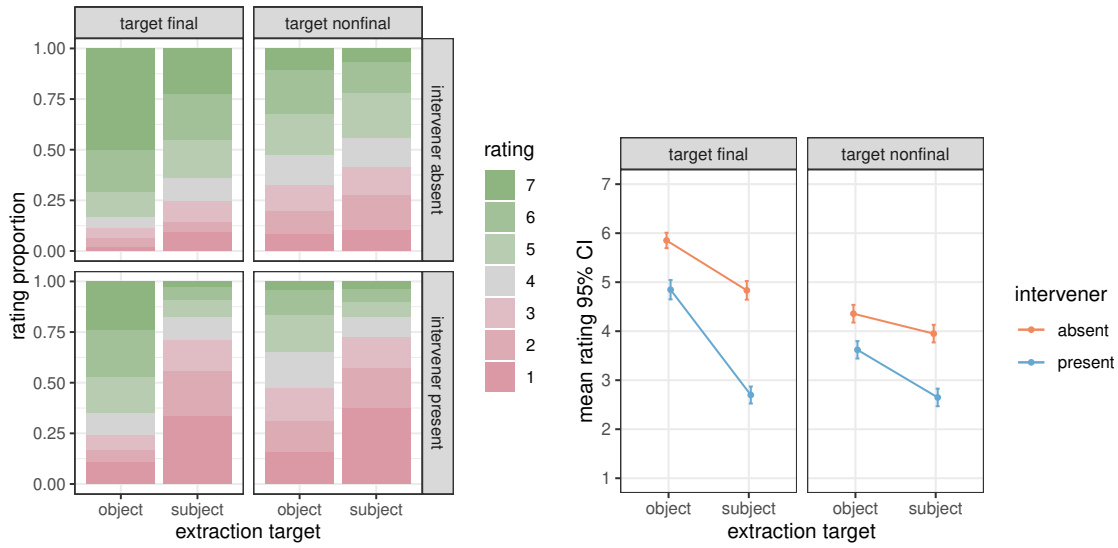


Figure 1: Results of the LBE experiment of Šimík and colleagues: proportion of individual ratings in each single condition (left) and mean ratings in each single condition (right)

randomization) and administered to participants via the L-Rex platform (most recent version: [Starschenko & Wierzba 2025](#)).

Prediction and statistical analysis: If Czech exhibits the expected subject–object asymmetry in LBE, we predict an interaction between EXTRACTION TARGET and INTERVENER. More specifically, we predict lower naturalness of extraction from subjects (as compared to extraction from objects), but only if the extraction is across an overt object intervener (as compared to the absence of such an intervener). We evaluate the prediction by fitting a Cumulative Link Mixed Model (function *clmm* of the package *ordinal*, [Christensen 2013](#), in R, [R Core Team 2024](#)). We use (EXTRACTION) TARGET (sum coded), INTERVENER (treatment coded; reference level: absent), (TARGET) POSITION (treatment coded; reference level: final), and the interaction between TARGET and INTERVENER as fixed factors and include random intercepts and slopes for participants and items as random effects.

Results: The results are visualized in Figure 1. We use two types of visualization: the proportion of individual ratings per condition (revealing potential anomalies like a bimodal distribution) and mean ratings per condition (making contrasts among individual conditions more visually salient). Our statistical model reveals a significant interaction between TARGET and INTERVENER ($z = 7.469, p < .001$): although the presence of an intervener negatively affects naturalness (main effect of INTERVENER: $z = -11.044, p < .001$), the effect is clearly stronger if the subextraction targets the subject than if it targets the object. In addition to the aforementioned effects, the model also reveals significant main effects of TARGET (extraction from a subject is less natural than extraction from an object; $z = 6.132, p < .001$) and of POSITION (extraction from a nonfinal position is less natural than extraction from the final position; $z = -9.729, p < .001$).

Discussion: The experiment of Šimík and colleagues clearly confirms the predicted subject–object asymmetry in Czech LBE: extraction from a subject across an object is perceived as significantly less natural than both extraction from an object across a subject and extraction from a subject without crossing an object. This shows that [Ko’s \(2007; 2014\)](#) analysis can be readily applied to Czech. We see that the naturalness of Czech LBE

is affected by some additional factors, not predicted by Ko's analysis or by any other analysis, as far as we know. We see that extraction from subjects is generally less natural than extraction from objects. This effect/tendency is visible in all conditions. We also see that the presence of an intervener causes a general decrease in naturalness, even in the cases of extractions from an object. We note that the latter effect is not due to a more complex / longer structure, as in conditions where an intervener was absent, it was "replaced" a temporal adverbial. We have no specific post-hoc explanation to offer for these effects, but hypothesize that they are a reflection of a non-grammatical factor, such as less frequent use of the affected conditions (see section 4 for more discussion on this). Finally, we note that the extraction from nonfinal constituents is consistently less natural than the extraction from final constituents (significant main effect of POSITION). This was first observed in Starke (2001) for extraction from subjects (Starke offers a tentative explanation embedded in his framework) and is discussed in passing for objects in Veselovská (2021). We do not have a satisfactory explanation of this phenomenon and can only speculate. One possibility is that it is a case of freezing – a ban on subextraction from moved constituents (Wexler & Culicover 1980; see also Hartmann et al. 2018 for more recent developments), which would supply a straightforward account of extractions from nonfinal objects (which have undoubtedly undergone movement in an SVO language like Czech; though cf. Haider & Szucsich 2022), but would necessitate auxiliary assumptions for subjects. More specifically, nonfinal (preverbal) subjects would have to be analyzed as evacuated from their base position in order to become "frozen" in the classical sense (but cf. Müller 2010). This in turn would entail obligatory movement of finite verbs in Czech out of vP (which is consistent with the analytical sketch in (2), but it is certainly not generally accepted; see e.g. Veselovská 1995). Another line of explanation might be sought in the domain of information structure (as briefly hinted at in Veselovská 2021), esp. in view of the fact that the clause-final position is characteristic for foci in Czech and foci, as opposed backgrounded constituents, are sometimes considered more transparent for extraction (e.g. Erteschik-Shir & Lappin 1979; Goldberg 2006). Whatever the ultimate explanation of the observed effect, it is good to keep in mind that it is arguably a phenomenon specific to Czech (and possibly Slovak; see Starke 2001), at least in the family of Slavic languages, as Russian (Bondarenko & Davis 2021) or Serbo-Croatian (Bašić 2004) are reported to readily allow for extraction from nonfinal constituents.

2.2 Humanness in grammar

Humanness – the property of denoting a human – is known to have an effect on word order (Müller 1999; Fanselow 2003; Titov 2012; 2017; Velnić 2019; Antonyuk 2022; a.o.). This is illustrated by the Russian examples (11) and (12). According to Titov (2017), example (11a) demonstrates that DO-IO (direct object before indirect object) is the basic order, as it is the least interpretively constrained: it can be used in all-focus contexts (response to 'What happened?') and places no constraints on the referential determinacy of the objects.⁶ In contrast, the IO-DO order in (11b) requires an interpretive license: in the absence of an information-structural license (in an all-focus context), the order is

⁶ Following Coppock & Beaver (2015), we call an NP determinate if it has a meaning of type *e*, i.e., it is not quantificational or predicative. In languages with articles, definite NPs are mostly determinate. In languages without articles, bare NPs can be both determinate and indeterminate (see e.g. Geist 2010). However, the issue has been subject to controversy: see Dayal 2011 for a determinate-only view and Heim 2011 or Seres & Borik 2021 for an indeterminate-only view; for an experimental perspective, supporting the latter position, see Šimík & Demian 2020).

licensed by determinacy: the scrambled (fronted) IO *osmotru* ‘inspection’ is interpreted as determinate and the DO *komnatu* ‘room’ as indeterminate.

(11) *Russian* (Titov 2017: 435–436)

Context: ‘What happened?’

- a. Ivan podverg komnatu osmotru.
Ivan subjected room.ACC inspection.DAT
‘Ivan subjected the/a room to the/a inspection.’
- b. Ivan podverg osmotru komnatu.
Ivan subjected inspection.DAT room.ACC
‘Ivan subjected a room to the inspection.’

What example (12) demonstrates, according to Titov (2017), is that humanness also provides a license for the IO-DO order. In (12b), the IO *agentu* ‘agent’ precedes the DO *pis’mo* ‘letter’ because the former denotes a human and the latter a thing. At the same time, there are no constraints imposed on the determinacy of both objects. In order to override the “basic” human–thing order, a determinacy-related license is needed, just as we saw in (11): in (12a), the thing-denoting ‘letter’ must be interpreted as determinate and the human-denoting ‘agent’ as indeterminate.

(12) *Russian* (Titov 2017: 433)

Context: ‘What happened?’

- a. Ivan peredal pis’mo agentu.
Ivan handed letter.ACC agent.DAT
‘Ivan handed the letter to an agent.’
- b. Ivan peredal agentu pis’mo.
Ivan handed agent.DAT letter.ACC
‘Ivan handed a/the letter to an/the agent.’

We note that the pattern is generally replicated in Czech (see e.g. Jasinskaja & Šimík *forthcoming*), the language under scrutiny in this article, though we should mention that we find the determinacy of bare NPs difficult to judge, esp. without any supporting context or scenario. Below we extend the observation to objects scrambling over subjects. The scenario and examples in (13) are set up in such a way that they do away with the issue determinacy: all the NPs involved are indeterminate.⁷ The examples demonstrate that when the humanness of both arguments is balanced – both human (HUM), (13b), or both nonhuman (NHUM), (13c) – the SVO order is preferred. When it is not, as in (13a) and (13d), it is humanness that determines the preferred order, which is highlighted by framing the critical patterns. Admittedly, the intuition is subtle, esp. in the critical case (13d), and would profit from more rigorous empirical support, but since our goal in this work is different, we assume the pattern in (13) as a working hypothesis about the data.

(13) *Czech* (authors’ own examples and intuition)

Scenario: A spring school festival is taking place in the schoolyard, featuring various activities, including an archery tryout. Suddenly, there’s a commotion. A teacher turns to her colleague and asks what happened. The colleague responds:

⁷ The indeterminacy is indicated in the meanings/translations by using indefinite NPs. The reason we characterize the meanings as “intended” is that in some cases, the word order might prompt one to accommodate determinacy, which in turn can ameliorate the infelicity. Consulted native speakers of Czech point out a preference for encoding the indeterminate reading by indefinite determiners like *nějaký* ‘some’. We keep the NPs bare for consistency with the Russian example discussed.

- | | | | |
|----|---|-----------------|----------------|
| a. | HUM S; NHUM O | | |
| | (i) Žačka rozbila okno. | | SVO / HUM-NHUM |
| | pupil.NOM.F broke.F window.ACC.N | | |
| | (ii) #Okno rozbila žačka. | OVS / NHUM-HUM | |
| | window.ACC.N broke.F pupil.NOM.F | | |
| | (Intended:) 'A pupil broke a window.' | | |
| b. | HUM S; HUM O | | |
| | (i) Žačka zasáhla návštěvníka. | SVO / HUM-HUM | |
| | pupil.NOM.F hit.F visitor.ACC.M | | |
| | (ii) #Návštěvníka zasáhla žačka. | OVS / HUM-HUM | |
| | visitor.ACC.M hit.F pupil.NOM.F | | |
| | (Intended:) 'A pupil hit a visitor.' | | |
| c. | NHUM S; NHUM O | | |
| | (i) Šíp rozbil okno. | SVO / NHUM-NHUM | |
| | arrow.NOM.M broke.M window.ACC.N | | |
| | (ii) #Okno rozbil šíp. | OVS / NHUM-NHUM | |
| | window.ACC.N broke.M arrow.NOM.M | | |
| | (Intended:) 'An arrow broke a window.' | | |
| d. | NHUM S; HUM O | | |
| | (i) #Šíp zasáhl žačku. | SVO / NHUM-HUM | |
| | arrow.NOM.M hit.M pupil.ACC.F | | |
| | (ii) Žačku zasáhl šíp. | OVS / HUM-NHUM | |
| | pupil.ACC.F hit.M arrow.NOM.M | | |
| | (Intended:) 'An arrow hit a pupil.' | | |

Even though humanness does not belong to the traditionally considered set of formal nominal features like case, number, or gender, mainly because it typically lacks inflectional morphological exponence, its relevance in grammatical systems has been acknowledged and well-documented. In Palauan, humanness is one of the features that triggers object agreement. Woolford (1999; 2000) analyzes this phenomenon in terms of object scrambling out of VP, following the spirit of Diesing's (1992) Mapping Hypothesis, i.e., the idea that certain types of NPs cannot be interpreted within the VP. The grammars of Mayan languages incorporate a restriction on subjects being at least as high on the animacy hierarchy as objects.⁸ Deal & Royer (online first) analyze this restriction as a case of *v*-subject agreement failure in situations where the object is featurally more specified than the subject, which – due to other grammatical circumstances in Mayan – leads to ungrammaticality. Another phenomenon where humanness is implicated is the case of differential object marking – different (case-)marking of objects depending on their semantic properties such as humanness (see, e.g., Bossong 1991; Aissen 2003; Fábregas 2013; Onea & Hole 2017; Haspelmath 2019).

Humanness also manifests itself in the grammar of Czech (and other Slavic languages).⁹ For instance, the possessive adjectival suffix *-ův* (masculine) / *-in* (feminine) only productively attaches to human-denoting nouns (Corbett 1987; Veselovská 1998). The strong form of pronouns (such as *jeho* 'him.STRONG' vs. *ho* 'him.CLITIC'), besides being necessarily contrastive, also presupposes reference to a human referent (Cardinaletti & Starke 1999). Finally, the nominative plural masculine animate suffix *-(ov)ě* also en-

⁸ For a discussion of the animacy hierarchy (in its simplest form [human > animate > inanimate]) and its grammatical relevance, see Silverstein (1976); Corbett (2000); Croft (2003); Siewierska (2004); Gardelle & Sorlin (2018); a.o.).

⁹ We are grateful to Markéta Ziková for a discussion about morphological reflexes of humanness in Czech.

codes humanness: *páni/pánové* ‘men.NOM.PL.M.ANIM.(HUM)’ or *sousedí/sousedé* ‘neighbors.NOM.PL.M.ANIM.(HUM)’ vs. *jeleni/*jelen(ov)é* ‘deer.NOM.PL.M.ANIM.(*HUM)’ or *hadí/*had(ov)é* ‘snakes.NOM.PL.M.ANIM.(*HUM)’.

While humanness specifically and animacy more generally have an undeniable effect on morphological and morphosyntactic processes (case-marking, agreement), its effect on scrambling as a narrowly syntactic operation has not been clearly established. While some researchers assume that humanness indeed affects syntactically encoded constituent ordering (esp. Titov 2012; 2017), others argue that the effect of humanness or animacy is more surface-oriented (Fanselow 2003), possibly related to processing efficiency (Seržant et al. in press), and/or subject to “weak” constraints, sometimes implemented in an optimality-theoretic fashion (Müller 1999).¹⁰

3 Humanness and left branch extraction

Having established the existence of the subject–object asymmetry in Czech LBE and the relevance of humanness for (Czech) word order and grammar more generally, we now move on to the core contribution of this paper. The research questions and hypotheses we aim to address are the following:

RQ1: Is the HUM-NHUM order just a surface preference or is it encoded in narrow syntax?

H1: The HUM-NHUM order is just a surface preference (e.g., Fanselow 2003).

H2: The HUM-NHUM order is encoded in narrow syntax.

RQ2: If the HUM-NHUM order is encoded in narrow syntax, what is the nature of the syntactic operation it triggers if the object is human and subject nonhuman?

H2a: The human object of a transitive predicate is base-generated in the complement of V (as per UTAH; Baker 1988) and undergoes movement (A-scrambling) to SpecvP, to become more prominent than the nonhuman subject [henceforth also called the UTAH-COMPATIBLE hypothesis].¹¹

H2b: The human object is base-generated in SpecVP and the nonhuman subject in the complement of V (Titov 2012) [henceforth also called the VP-BASED SWAPPING hypothesis]¹²

H2c: The human object (reconceptualized as an experiencer) is base-generated in SpecvP and the nonhuman subject (reconceptualized as theme) in the complement of V [henceforth called the vP-BASED SWAPPING hypothesis]¹³

¹⁰ Optimality theory is sometimes also invoked in morphosyntactic work on humanness – see Woolford (2000); Aissen (2003).

¹¹ This hypothesis is compatible with a standard movement-approach to (Slavic) A-scrambling, assumed e.g. in Bailyn (2004), Biskup (2011), or Antonyuk (2015). It is rarely assumed though that humanness (or animacy) can trigger such a movement; a recent exception is Antonyuk (2022), who, however, only discusses scrambling across internal arguments. In addition, individual analyses differ in important technical details, making potentially different predictions. Bailyn (2004), for instance, assumes object movement to SpecIP, without stopping at the edge of vP, which would predict a different empirical pattern than assumed here. We do not go into detail about the individual proposals for reasons of space.

¹² Titov (2012: 120ff.) explicitly argues that the object must be generated in SpecVP, not SpecvP in this type of configuration.

¹³ This analysis creates a configuration akin to the one entertained for object-experiencer predicates (Belletti & Rizzi 1988), treated here experimentally in section 4.5. We note that under the assumption that humanness affects theta roles, this analysis is also “UTAH-compatible”.

We address these questions and hypotheses by investigating the interaction between argument humanness manipulations and LBE. The empirical pattern under consideration is provided in (14) (a schematic version of (1)), along with a summary of the predictions of the individual hypotheses (✓ = predicted to be grammatical; ✗ = predicted to be ungrammatical). If humanness only affects surface preferences (H1), we do not expect any strong effect of humanness manipulations on the subject–object asymmetry in LBE: extraction from a human object across a nonhuman subject is expected to be grammatical, with at most a slightly decreased naturalness due to the dispreferred NHUM-HUM order. If, on the other hand, humanness affects the syntactic position of arguments, i.e., if it is encoded in narrow syntax, we expect it to have a clearly detectable effect on naturalness. Moreover, the effect will be different depending on the type of analysis adopted. The UTAH-compatible hypothesis H2a is the most restrictive one in that it rules out not only extractions from a subject across an object, but also the extraction from a human object across a nonhuman subject. The VP-based swapping hypothesis H2b (Titov 2012) is the most permissive one in that it predicts extraction from a subject across an object to be grammatical as long as the subject is a nonhuman and the object a human. Finally, the vP-based swapping hypothesis H2c predicts subextraction to only be possible from nonhuman arguments across human arguments (if humanness of both arguments is imbalanced, which holds of all four cases in (14)).

(14)	Pattern under consideration	H1	H2a	H2b	H2c
a.	which.ACC cook.NOM poisoned sauce.ACC WH HUM S NHUM O	✓	✓	✓	✓
b.	which.ACC sauce.NOM poisoned cook.ACC WH NHUM S HUM O	✓	✗	✓	✗
c.	which.NOM cook.ACC poisoned sauce.NOM WH HUM O NHUM S	✗	✗	✓	✓
d.	which.NOM sauce.ACC poisoned cook.NOM WH NHUM O HUM S	✗	✗	✗	✗

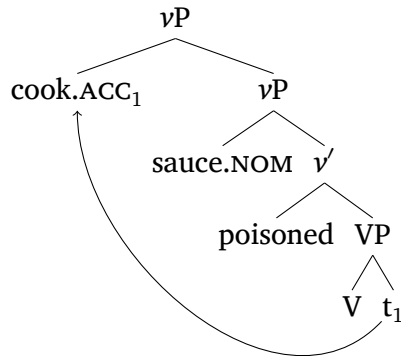
Let us now unpack the expectations in more detail. We will be quick with H1, as its expectations do not differ dramatically from the baseline subject–object asymmetry described in section 2.1: extraction from objects across subjects (14a)/(14b) are predicted to be grammatical and extraction from subjects across objects (14c)/(14d) ungrammatical, irrespective of the humanness of individual arguments. It is reasonable to expect a slight decrease in naturalness in (14b), where the order does not align with the assumed surface preference. However, this decrease should be smaller than in conditions (14c)/(14d). The situation is different if we assume that the humanness of arguments affects their narrowly syntactic behavior, which we turn to now.

Let us first consider the UTAH-compatible hypothesis (H2a), in which the object is base-generated in the complement of V, where it receives its patient theta-role, regardless of its humanness. If the object is human and the subject nonhuman, however, the object is expected to scramble over the subject to a higher SpecvP, as illustrated in (15). The idea is that the scrambling is obligatory, unless there is a stronger factor requiring the subject to stay in a more prominent position, such as the determinacy or topicality of the subject and, by contrast, indeterminacy or focality of the object (following the spirit of multifactorial approaches to scrambling, such as Müller 1999 or Titov 2012).¹⁴ The

¹⁴ We put these potentially confounding factors aside for purposes of our main experiment – Experiment 1 (section 4.2) –, assuming that they do not play a significant role in the absence of any supporting context,

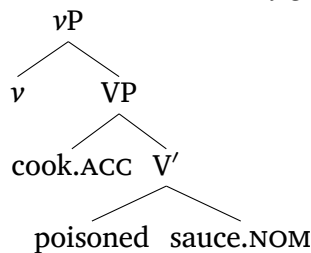
configuration makes evident that extraction from a nonhuman subject across a human object (14c) will fail for the same reasons for which extraction from subjects across objects fails more generally; see section 2.1. More importantly, extraction from a human object across a nonhuman subject (14b) also fails, as it would necessarily involve scrambling the nonhuman subject across the human object, which, however, would constitute a violation of the ban on spec-to-spec movement.

- (15) The UTAH-compatible hypothesis (H2a)
Pre-subextraction configuration of NHUM S + HUM O (14b)/(14c)



We now move on to the VP-based swapping hypothesis (H2b) proposed by Titov (2012), who in turn builds on Neeleman & van de Koot (2002). The baseline configuration of NHUM S + HUM O is provided in (16): the non-canonical (non)humanness of both arguments causes the object to be more structurally prominent and independent assumptions of the modified theta-theory force both arguments to be generated within the VP. This makes the phase-edge unused in the baseline configuration, which in turn significantly opens up subextraction possibilities. Consider first the case of extraction from a human object across a nonhuman subject (14b). This is achieved by scrambling the subject across the object, targeting SpecvP, followed by the subextraction of ‘which’ from the object, to a higher SpecvP. Both movements are licit, as they are movements from VP to SpecvP, inducing no spec-to-spec violation. Moreover, the linearization statements at vP are compatible with those at CP. The extraction from a nonhuman subject across a human object (14c) is likewise unproblematic: ‘which’ is subextracted from the VP-internal subject to SpecvP and later moves to SpecCP.

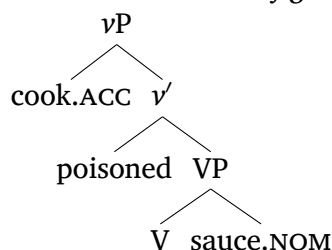
- (16) The VP-based swapping hypothesis (H2b)
Pre-subextraction configuration of NHUM S + HUM O (14b)/(14c)



which is arguably required for both determinacy and information structure (topicality or focality). We are aware of the risk that these factors are manipulated in the minds of our participants (and thus beyond our control), relying on (presupposition) accommodation (see e.g. Romoli & Sauerland 2017), but have decided against manipulating them explicitly in Experiment 1 in the interest of keeping the design simple. However, we come back to the issue of determinacy in Experiment 3 (section 4.4).

The *vP*-based swapping hypothesis (H2c) has not been seriously considered in the literature so far. We include it here mainly for the sake of completeness and comparison with standard analyses of object-experiencer predicates (see section 4.5), with which it bears some resemblance. As illustrated in (17), the human object is base-generated in Spec*vP*, as is standardly assumed for experiencer objects. The nonhuman subject is then treated as the generic theme argument and is generated in the complement of *V*. This configuration is thus a mirror image of the one assumed by H1 and effectively reverses the subject–object asymmetry (or the case of NHUM S + HUM O). In particular, it creates the expectation that extraction from a nonhuman subject across a human object is grammatical, but extraction from a human object across a nonhuman subject is not, as it would involve the scrambling of the nonhuman subject to a higher Spec*vP*, excluding any subextraction from the human object (as a result of a linearization conflict or a spec-to-spec ban violation).

- (17) The *vP*-based swapping hypothesis (H2c)
Pre-subextraction configuration of NHUM S + HUM O (14b)/(14c)



We are now in a position to test the predictions experimentally.

4 Experiments

We report a number of experiments which address – directly or indirectly – the research questions stated in section 3. Experiment 1 addresses these questions directly and Experiments 2–4 provide some additional corroborating evidence.¹⁵

The motivation to rely on experimental data rather than intuitive judgments at an individual level is that the pattern at hand combines a number of properties, which cumulatively add up to the overall complexity and oddity of the construction (and arguably also infrequency, though we have not performed any corpus analysis): left branch extraction, extraction across another nominal phrase (see section 2.1), the use of multiple bare NPs within one clause, and reversible transitive predicates compatible with human and non-human subjects and objects. In this setting of an overall decreased naturalness (caused by processing factors), providing judgments about syntax feels difficult and unreliable.¹⁶

4.1 General experimental setup

All reported experiments were part of a single experimental setup (also called “study” below) in the sense that their stimuli were mixed on the same lists and functioned as mutual fillers / distractors. This affords direct post hoc comparisons of specific conditions across experiments. Table 1 provides an overview of the LBE experiments reported in the

¹⁵ All experimental materials, results, and analyses can be found at https://osf.io/8sjyu/?view_only=c9ecf634a8b5445085feb93f5045db77.

¹⁶ For relevant methodological discussion about judgments of acceptability, naturalness, or grammaticality, see, e.g., Keller (2000); Featherston (2005).

	Experiment name	§	Number of items
1	Humanness and LBE	4.2	24
2	Balanced humanness	4.3	8
3	Determinate interveners	4.4	8
4	Object experienter predicates	4.5	16
	other distractors		54
	total number of items		110

Table 1: Experiments and distractors in the experimental setup and number of items/stimuli

following (numbered and referenced by section), the number of items in each experiment, as well as the number of other distractor and control items (used, among other things, to verify the quality of the participants). Experiment 1, testing the hypotheses laid out in section 3, is the core contribution of this paper and is therefore powered accordingly: with 4 conditions and with 24 items, each unique condition receives 6 ratings from each participant. The other experiments are powered less, involving between 1 and 4 ratings per condition from each participant, and could thus be considered pilot experiments inviting more solid experimentation.

Participants: 174 participants completed the experimental study. All were self-identified native speakers of Czech and university students of a variety of study programs, selected at random from among the voluntary members of the LABELS lab participant pool affiliated with the Faculty of Arts of Charles University and the Institute of Psychology at the Czech Academy of Sciences. Successful participation was counted towards receiving university credits.

Administration and procedure: The experimental study was prepared (including Latin Square distribution and pseudo-randomization) and administered online via the L-Rex platform (Starschenko & Wierzba 2025) in March 2025. The participants were instructed to carry out the study individually, on their own computer, and in a calm environment. The participants were asked to rate the naturalness of 110 sentential stimuli, which they were exposed to one-by-one, on a scale from 1 (completely unnatural) to 7 (completely natural). The task itself was preceded by informed consent and simple instructions, which included two examples – one completely unnatural (ungrammatical) sentence and one completely natural sentence. There was no time pressure; most participants took 20 to 30 minutes to complete the study.

Inclusion criterion and analysis: We checked the participant quality based on 8 pre-selected stimuli from the distractors – 4 completely unnatural and 4 completely natural (as judged by the authors – native speakers of Czech). The inclusion criterion was to provide an expected rating in at least 6 of the 8 control stimuli. An expected rating for natural stimuli was 5 or better; for unnatural stimuli 3 or worse. 147 participants have passed the criterion and their data have been included in all the analyses reported below. All statistical analyses have been performed in R (R Core Team 2024) by fitting Cumulative Link Mixed Models (function *clmm* of the package *ordinal*; Christensen 2013), one for each experiment. Following Bates et al. (2015), we applied the most complex models possible, including random intercepts and slopes for participants and items, whenever the model converged and yielded meaningful results. In some cases, we applied post-hoc pairwise comparisons using the function/package *emmeans* (Lenth et al. 2025).

4.2 Experiment 1: Humanness and left branch extraction

The goal of Experiment 1 is to address the questions and hypotheses laid out in section 3, namely whether humanness affects narrowly syntactic configurations and, if it does, then in which way exactly. The way we approach the questions is by looking at left branch extraction and how its naturalness is affected by manipulating the humanness of both arguments of a transitive predicate.

4.2.1 Design and materials

The design is fully correlated with the pattern in (14), representative of example (1), which in turn is repeated below (without grammaticality annotations) to exemplify one of the 24 items constructed for Exp. 1. We manipulated two factors in a 2×2 design, both within participants and within items: CONSTITUENT ORDER (subj-obj vs. obj-subj) and HUMANNES ORDER (h(um)-nh(um) vs. nh(um)-h(um)), in order to provide the pattern needed to test our hypotheses. Each item contains a short matrix clause followed by an embedded question introduced by ‘which’ extracted from the question’s final constituent (motivated empirically above). The reasons for choosing wh-extraction over extractions of other types of elements (demonstratives, adjectives) were detailed in section 2.1. The reason why we chose embedded rather than matrix questions was better alignment with the design of Šimík and colleagues (see section 2.1) and partly letting the participants judge the naturalness of only one type of clause throughout the study, that is, a statement. The embedded question consists of 4 words – a wh-word, an argument (subject or object), a transitive verb, and the other argument (remnant subject or object). The NPs used in the argument position were selected in such a way that a parse where the wh-word forms a constituent with the immediately following NP is impossible. The verbs were carefully selected so that they allow (i) the reversibility of both arguments and (ii) human and nonhuman entities in both the external and internal argument slot; both properties were required by our within-items design.¹⁷

- (18) Detektiv na začátku vyšetřování tušil, ...
 detective at beginning investigation had.idea
 ‘At the beginning of the investigation, the detective had an idea...’
- a. *kterou kuchařka otrávila omáčku.* [subj-obj, h-nh]
 which.ACC cook.NOM poisoned sauce.ACC
 (Intended:) ‘which sauce a/the cook poisoned.’
 - b. *kterou omáčka otrávila kuchařku.* [subj-obj, nh-h]
 which.ACC sauce.NOM poisoned cook.ACC
 (Intended:) ‘which cook a/the sauce poisoned.’
 - c. *která kuchařku otrávila omáčka.* [obj-subj, h-nh]
 which.NOM cook.ACC poisoned sauce.NOM
 (Intended:) ‘which sauce poisoned a/the cook.’
 - d. *která omáčku otrávila kuchařka.* [obj-subj, nh-h]
 which.NOM sauce.ACC poisoned cook.NOM
 (Intended:) ‘which cook poisoned a/the sauce.’

¹⁷ We have profited from the VALLEX dictionary (Lopatková et al. 2016) for purposes of item construction. Still, the task has proved an uneasy one, which has complicated the creation of naturally sounding stimuli and which in turn might have contributed to the overall low naturalness of the experimental stimuli.

4.2.2 Hypotheses and predictions

We consider four hypotheses, motivated and described in more detail in section 3: the hypothesis that humanness has no direct effect on syntactic configurations (H1), the hypothesis that human objects obligatorily scramble across nonhuman subjects (UTAH-compatible hypothesis; H2a), the hypothesis that human objects are base-generated in a more prominent position (namely SpecVP) than nonhuman subjects (complement of V) (VP-based swapping hypothesis; H2b), and finally the hypothesis that human objects are construed as experiencers, base-generated in SpecvP, in the presence of nonhuman subjects, which are construed as themes, generated in the complement of V (vP-based swapping hypothesis; H2c).

The predictions of the individual hypotheses are repeated below with the experimental conditions explicitly specified. In what follows we formulate them in terms of the expected outcomes of the statistical models. H1 predicts a main effect of CONSTITUENT ORDER, in accordance with the general subject–object subextraction asymmetry: extraction from objects across subjects are predicted to be licit, from subjects across objects not (see section 2.1). H2a predicts an interaction between CONSTITUENT ORDER and HUMANNESS ORDER: besides the expected unnaturalness of extraction from a subject across an object, extraction from an object across a subject should be unnatural in a non-canonical humanness alignment, that is, when the object denotes a human and the subject a nonhuman (19b). The “most permissive” H2b also predicts an interaction between the two factors, but with a very different profile: it expects the subject–object asymmetry to apply, but only in the canonical humanness distribution; the ban on extraction from subjects across objects is expected to be lifted in the non-canonical human alignment – with human objects and nonhuman subjects (19c). Finally, H2c predicts a main effect of HUMANNESS ORDER: extraction from nonhuman constituents across human constituents should be more natural than the reverse pattern.

(19)	<i>Pattern under consideration</i>	H1	H2a	H2b	H2c
a.	[subj-obj, h-nh] which.ACC cook.NOM poisoned sauce.ACC WH HUM S NHUM O	✓	✓	✓	✓
b.	[subj-obj, nh-h] which.ACC sauce.NOM poisoned cook.ACC WH NHUM S HUM O	✓	✗	✓	✗
c.	[obj-subj, h-nh] which.NOM cook.ACC poisoned sauce.NOM WH HUM O NHUM S	✗	✗	✓	✓
d.	[obj-subj, nh-h] which.NOM sauce.ACC poisoned cook.NOM WH NHUM O HUM S	✗	✗	✗	✗

4.2.3 Results and discussion

Results: The results of Experiment 1 are visualized in Figure 2, using the same visualization strategy introduced in section 2.1: one graph shows the relative proportion of the individual ratings per condition and the other shows mean ratings supported by so-called interaction lines, affording a more intuitive comparison of the individual conditions. One striking finding is the overall low naturalness of all the stimuli. Even the best-rated con-

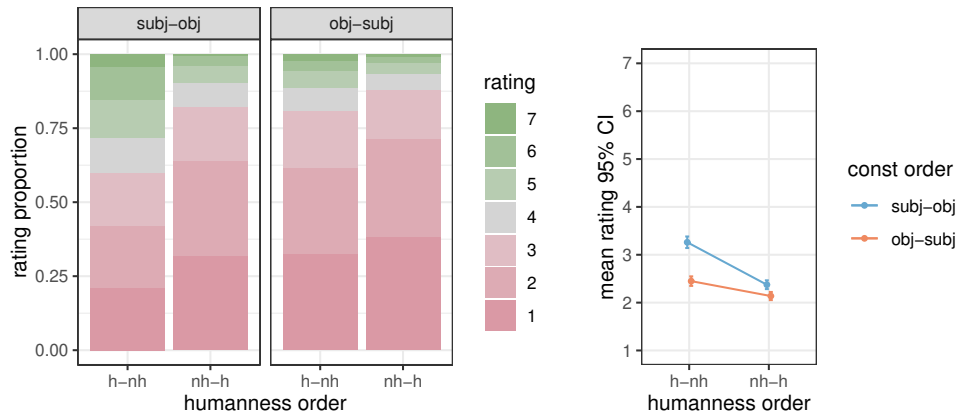


Figure 2: Results of Experiment 1: proportion of individual ratings in each single condition (left) and mean ratings in each single condition (right)

dition, namely subj-obj, h-nh, has a median rating of 3, modus 2, and mean 3.3. We return to this point below in the discussion.

Our statistical model included CONSTITUENT ORDER, HUMANNES ORDER, and their interaction as fixed effects and random intercepts and slopes for both participants and items as random effects. Both CONSTITUENT ORDER and HUMANNES ORDER were treatment-coded, with reference levels set to subj-obj and h-nh, respectively. The model revealed a significant interaction between the two factors ($z = 2.388, p = .017$), confirming the visually salient pattern: extraction from objects across subjects is more natural, but this effect is stronger if the object is nonhuman and the subject human.¹⁸ The model also revealed a main effect of both CONSTITUENT ORDER and HUMANNES ORDER in the model: extraction from subjects across objects is less natural ($z = -5.353, p < .001$) and extraction from human across nonhuman constituents is less natural ($z = -5.846, p < .001$). A post-hoc pairwise comparison computed by the *emmeans* function (Lenth et al. 2025) confirms that the obj-subj, nh-h is significantly less natural than obj-subj, h-nh (simple effect of HUMANNES ORDER; $z = 3.203, p < 0.01$) and than subj, obj, nh-h (simple effect of CONSTITUENT ORDER; $z = 3.111, p = 0.01$).

Discussion: The results of Experiment 1 are in line with the predictions of UTAH-compatible hypothesis (H2a). This hypothesis follows from the idea that patients are base-generated in the complement of V and agents in SpecvP, irrespective of their humanness. Humanness does have an effect on narrow syntax, however, forcing a human object to scramble over a nonhuman subject, effectively rendering extraction from human objects across nonhuman subjects (condition (19b)) ungrammatical, as detailed in section 4.2. We note that there is also a confounding interpretation consistent with H1: a similar result would be expected if there is (i) the vanilla subject-object asymmetry in LBE (confirmed independently in section 2.1) and in addition (ii) a penalty for nonhuman subjects of transitive predicates. The latter would push the naturalness of condition subj-obj, nh-h down independently of the assumptions adopted under H2a. We consider this interpretation unlikely in view of the fact that an analogous penalty is not observed in the obj-subj, h-nh condition, which exhibits a nonhuman subject in addition to the prob-

¹⁸ Even though the interaction is statistically not particularly robust, we note that this is already the second replication of the same result; the first experiment with the same design (and with very similar items) has been published in the BA thesis of the first author (Kořánová 2023). The successful first replication was achieved in a less powered pilot of this experiment run in January 2025.

lematic extraction from subjects across objects. Moreover, Experiments 2 and 3 show that there is no general naturalness penalty of nonhuman subjects.

What is also worth discussing is that we see not just the interaction predicted by the UTAH-compatible hypothesis (H2a), but also the singling out of one of the conditions, as evident from the post-hoc pairwise test. More specifically, the conditions that are ungrammatical – by hypothesis – are not all unnatural to the same degree: condition obj-subj, nh-h is the least natural one of them. While we have no definitive explanation of this unpredicted phenomenon, we conjecture that it is a spurious processing effect: the least natural condition combines two phenomena that are infrequent and arguably difficult to process even outside of an LBE setting.¹⁹

Finally, we would like to discuss briefly the overall low naturalness rating in this experiment. As we already laid out in the introduction to section 4, an overall low naturalness of stimuli in this type of experiment is to be expected given the cumulation of factors complicating the processing of the whole structure, including LBE itself, the use of multiple bare NPs, or reversible transitive predicates. The fact that our experiment reveals a meaningful significant interaction despite the assumed processing complications demonstrates the strength of the applied methodology. As has been argued by Featherston (2005), among others, the violation of a constraint is expected to be detectable independently of the overall naturalness of the construction tested. It is also worth noting that the “strength” of the penalty, understood in terms of the decrease in absolute naturalness on the 7-point scale, is comparable to the one observed in the experiment of Šimík and colleagues reported in section 2.1, particularly in their nonfinal condition. In this condition, affected by the unnaturalness of extraction from a nonfinal constituent, the subject–object asymmetry leads to decrease by about 1 point on the scale, similarly to what we see in Experiment 1. Furthermore, we will see that the generally low naturalness is also observed in Experiments 2–4, corroborating our interpretation of the results.

4.3 Experiment 2: Balanced humanness

In Experiment 2, we aim to address the question whether the subject–object asymmetry in LBE is observed not just in cases with canonical humanness distribution (HUM S + NHUM O), but also in cases where the humanness of both arguments is balanced (HUM S + HUM O and NHUM S + NHUM O). The experiment will also address the confounding interpretation of the results of Experiment 1, where the relative unnaturalness of the subj-obj, nh-h condition could be attributed to the non-canonical nonhuman denotation of the subject.

4.3.1 Design and materials

We used a 2×2 within-participants within-items design, manipulating CONSTITUENT ORDER (subj-obj vs. obj-subj) and the HUMANNESSESS of the arguments (hum vs. nhum). An example of an item is provided in (20). The construction of all the 8 items followed the same tenets that were detailed in the design of Experiment 1.

- (20) Na záběru z dopravní kamery bylo vidět, ...
 on footage from traffic camera was see.INF
 ‘It was possible to see on the footage from the traffic camera...’

¹⁹ It is not without relevance that this is the first time we observe this effect; it was absent in both previous runs of this experiment, which were less powered and which only exhibited the predicted interaction.

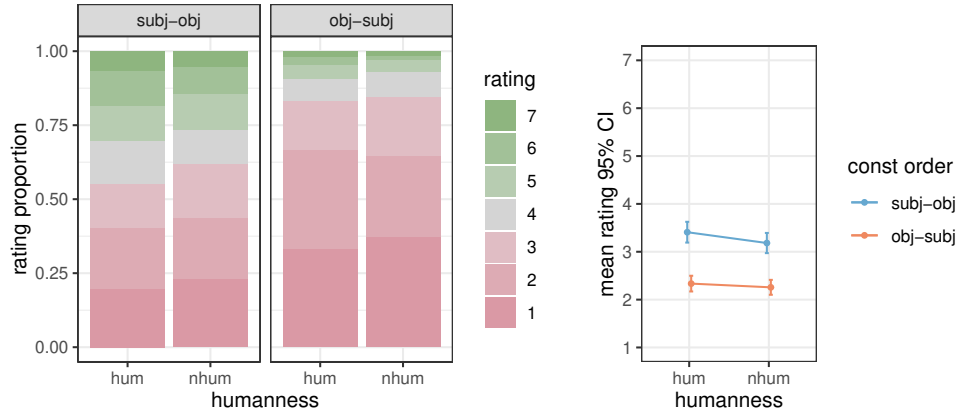


Figure 3: Results of Experiment 2: proportion of individual ratings in each single condition (left) and mean ratings in each single condition (right)

- a. *kterou řidička objela policistku.* [subj-obj, hum]
 which.ACC driver.NOM drove.around policewoman.ACC
 ‘which policewoman a/the driver drove around.’
- b. *která policistku objela řidička.* [obj-subj, hum]
 which.NOM policewoman.ACC drove.around driver.NOM
 ‘which driver drove around a/the policewoman.’
- c. *které motorka objela město.* [subj-obj, nhum]
 which.ACC motorbike.NOM drove.around town.ACC
 ‘which town a/the motorcycle drove around.’
- d. *která město objela motorka.* [obj-subj, nhum]
 which.NOM town.ACC drove.around motorbike.NOM
 ‘which motorbike drove around a/the town.’

4.3.2 Prediction

All the accounts considered in this paper make the same prediction: when humanness of both arguments is balanced, we should see the same subject–object asymmetry observed in the literature and also found in experimental results so far.

4.3.3 Results and discussion

Results: The results are visualized in Figure 3. Besides the low naturalness values familiar from Experiment 1, we see a clear effect of CONSTITUENT ORDER: extraction from a subject across an objects is less natural than extraction from an object across a subject. This effect is confirmed by our statistical model, which included CONSTITUENT ORDER (treatment-coded; reference level: subj-obj), HUMANNES (sum-coded), and their interaction as fixed effects and random intercepts and slopes for participants and items as random effects. The main effect of CONSTITUENT ORDER is significant ($z = -6.799, p < .001$) and there is no effect of HUMANNES ($z = 1.024, p = .31$) and no significant interaction between both factors ($z = -0.589, p = .56$).

Discussion: The results confirm the expectation that when humanness of both arguments is balanced, the baseline configuration is determined by constituent function (subject/agent vs. object/patient). The experiment is yet another confirmation of the subject–object asymmetry in LBE. At the same time, it shows that there is no general naturalness

penalty of nonhuman subjects, which we briefly contemplated in the discussion of the results of Experiment 1.

4.4 Experiment 3: Determinate interveners

The aim of Experiment 3 is to see whether the effect of humanness is overridden by determinacy, which is considered a stronger factor than humanness (or more generally animacy) in hierarchical multifactorial approaches to word order (Müller 1999; Titov 2012). Consider example (21). Example (21a) represents the critical condition of Experiment 1. Assuming the UTAH-compatible hypothesis, which was supported by the results of Experiment 1, the human object must scramble over the nonhuman subject. As a result, both the object and subject are positioned at the edge of the vP phase, which in turn renders LBE (from object and subject alike) impossible. Example (21b) differs minimally in that the nonhuman subject is determined by a prenominal demonstrative, making it unambiguously determinate (cf. section 4.4.1 for a proviso). If determinacy takes priority over humanness, then (21b) involves no scrambling of the human object in front of the nonhuman subject; it is the configuration DET-INDET (corresponding to SUBJ-OBJ in (21b)) that takes precedence over HUM-NHUM, thus rendering object scrambling unnecessary and therefore impossible (see esp. Titov 2012 and section 2.2 for discussion).

- (21) a. *... *kterou omáčka otrávila kuchařku.*
 which.ACC sauce.NOM poisoned cook.ACC
 Intended: ‘... which cook a/the sauce poisoned.’
 b. ... *kterou ta omáčka otrávila kuchařku.*
 which.ACC DEM sauce.NOM poisoned cook.ACC
 (Intended:) ‘... which cook this/that/the sauce poisoned.’

4.4.1 Design and materials

We used a 2×2 design, manipulating (within participants and items) CONSTITUENT ORDER (subj-obj vs. obj-subj) and SUBJECT HUMANNESS (shum vs. snhum). The humanness of the object always contrasted with that of the subject, yielding an imbalanced pattern, as in Experiment 1. As is evident from the example item (22)/(23), the intervener – whether the subject (in extractions from an object) or the object (in extractions from a subject) – was always determinate. More specifically, it was a demonstrative NP, referring anaphorically to an antecedent introduced in the matrix clause. For this reason, there were two versions of the introductory clause – (22) vs. (23) –, correlating with the intervener used. The reason why we relied on a contextual cue (anaphoric reference) rather than just a formal cue (demonstrative NP) in controlling the determinacy of the intervener was that in Czech, the use of a demonstrative alone does not guarantee the determinacy of the NP, as argued in Šimík (2016). The critical condition is (23a) (cf. (21)), which involves extraction from a human object across a determinate nonhuman subject.

- (22) Po odchodu asistentky z pódia kouzelnického představení jsem zapomněl,
 after departure assistant.GEN from stage magic show AUX forgot
 ...
 ‘After the/an assistant left the stage of the magic show...’

- a. *které* *ta* *asistentka* *ukázala zrcadlo.* [subj-obj, shum]
 which.ACC DEM assistant.NOM showed mirror.ACC
 ‘which mirror the assistant showed.’
- b. *které* *tu* *asistentku* *ukázalo zrcadlo.* [obj-subj, shum]
 which.NOM DEM assistant.ACC showed mirror.NOM
 ‘which mirror showed the assistant.’
- (23) Po odjezdu zrcadla z pódia kouzelnického představení jsem zapomněl,
 after departure mirror.GEN from stage magic show AUX forgot
 ...
- ‘After the/a mirror left the stage of the magic show...’
- a. *kterou* *to* *zrcadlo* *ukázalo asistentku.* [subj-obj, snhum]
 which.ACC DEM mirror.NOM showed assistant.ACC
 ‘which assistant the mirror showed.’
- b. *která* *to* *zrcadlo* *ukázala asistentka.* [obj-subj, snhum]
 which.NOM DEM mirror.ACC showed assistant.NOM
 ‘which assistant showed the mirror.’

We constructed 8 items like (22)/(23). Each item had 4 more conditions, in fact, stemming from the fact that we manipulated an additional factor, namely the categorical nature of the intervener, which was either an NP, or a pronoun (or a *pro*, if it was a subject). We only report on the NP results here because they have direct relevance to our research questions.

4.4.2 Predictions

If determinacy overrides the effect of humanness (Müller 1999; Titov 2012), we expect the critical subj-obj, snhum condition to be as natural as the baseline subj-obj, shum condition. This means that the naturalness decrease witnessed in the parallel condition in Experiment 1 (that is, subj-obj, nh-h) should not be observed. Speaking in terms of our statistical analysis, all we expect to find is a main effect of CONSTITUENT ORDER. No interaction is expected between CONSTITUENT ORDER and SUBJECT HUMANNES. For completeness, let us note that just that interaction would be expected if determinacy played no role; in that case, the results should mirror those of Experiment 1.

4.4.3 Results and discussion

Results: The results of Experiment 3 are visualized in Figure 4. The statistical model we applied included CONSTITUENT ORDER (treatment-coded; reference level: subj-obj), SUBJECT HUMANNES ((treatment-coded; reference level: shum), and their interaction as fixed effects. Since there was a smaller number of data points (one participant rated each condition only once), the random-effects structure was simplified compared to the models applied above and involved random intercepts for participants and items and random slopes for participants. The model reveals a main effect of CONSTITUENT ORDER ($z = -5.271, p < .001$), but no other statistically significant effects (effect of SUBJECT HUMANNES: $z = -1.012, p = 0.31$; interaction between both factors: $z = 0.585, p = 0.56$).

Discussion: The results clearly corroborate the expectation of what we have called multifactorial hierarchical approaches to scrambling: human objects must scramble over

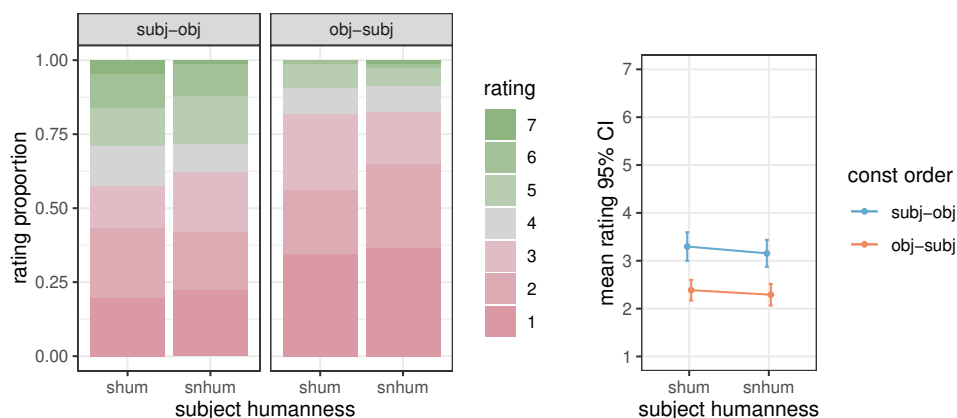


Figure 4: Results of Experiment 3: proportion of individual ratings in each single condition (left) and mean ratings in each single condition (right)

nonhuman subjects, but if the nonhuman subject is determinate (and the human object indeterminate), scrambling does not take place. If the scrambling does not take place, LBE is free to target the object. Experiment 3 shows that this expectation is borne out.

4.5 Experiment 4: Object experiencer predicates and left branch extraction

In our previous experiments, we have seen ample evidence for the subject–object asymmetry in left branch extraction (Saito 1985; Ko 2007; 2014; Bondarenko & Davis 2021; a.o.). The analysis of the asymmetry (due to Ko 2007) crucially relies on the idea that subjects (or, more precisely, agents) are base-generated in SpecvP and objects (patients) in the complement of V. In Experiment 4, we put the assumption under scrutiny by looking at predicates, where the base-generation configuration is arguably opposite, namely OBJECT-EXPERIENCER PREDICATES, a special category of so-called psych (state) predicates. Relevant examples from Czech (drawing from our stimuli) are provided in (24), (24a) involving a dative experiencer and (24b) an accusative experiencer.

(24) Czech

- a. Porotkyni se zalíbila zpěvačka / písnička.
judge.DAT REFL liked singer.NOM song.NOM
'A/The judge liked a/the singer/song.'
- b. Dirigenta zaujala skladatelka / skladba.
conductor.ACC draw.one's.attention composer.NOM composition.NOM
'A/The composer/composition drew a/the conductor's attention.'

Since Belletti & Rizzi (1988), these predicates have been commonly analyzed as fundamentally different from transitive predicates, in that the object is base-generated in a more prominent position – in today's terms in SpecvP (relying on a dedicated experiencer little *v*; cf. the standard agentive *v*) – than the subject, which is analyzed as bearing the default theta role called theme and is generated in the complement of V. Object-experiencer predicates thus represent a case which is arguably properly analyzed in accordance with the vP-based swapping hypothesis; see the structure in (17).²⁰ An argument for such an analysis comes from word order: object experiencers canonically

²⁰ We are aware that this is not the only analytical option and that there are competing proposals making potentially different predictions, but we leave them aside here for space reasons. The relevant literature

precede subject themes, even if the subject is human. Another argument for an underlying analysis that is qualitatively different from that of transitive predicates is that psych verbs cannot be passivized. On the other hand, it still makes sense to call the nominative theme argument a subject (and the dative/accusative experiencer argument an object), because the nominative (unlike the dative/accusative) generally retains its capacity to bind reflexive anaphors.

4.5.1 Design and materials

In a 2×2 design, we manipulated CONSTITUENT ORDER (subj-obj vs. obj-subj), SUBJECT HUMANNESS (shum vs. snhum), both within participants and items, and the CASE of the experiencer (dative vs. accusative), the latter within participants, but between items.²¹ The experiencer is invariantly a human in our design, as sentience appears to be an obligatory property of objects of psych predicates. Thus, there is no option to make use of reversibility, as we did in the previous experiments.²² An example item (from the dative-experiencer set) is provided in (25). We constructed 8 items for each level of CASE, i.e., 16 items in total.²³

- (25) Publikum ve studiu pěvecké soutěže odhadovalo, ...
 audience in studio singing competition guessed
 ‘The audience in a studio during a singing competition guessed...’
- a. *které se zpěvačka zalíbí porotkyni.* [subj-obj, shum]
 which.DAT REFL singer.NOM likes judge.DAT
 ‘which judge would like a/the singer.’
- b. *která se porotkyni zalíbí zpěvačka.* [obj-subj, shum]
 which.NOM REFL judge.DAT likes judge.NOM
 ‘which singer a/the judge would like.’
- c. *které se písnička zalíbí porotkyni.* [subj-obj, snhum]
 which.DAT REFL song.NOM likes judge.DAT
 ‘which judge would like a/the song.’
- d. *která se porotkyni zalíbí písnička.* [obj-subj, snhum]
 which.NOM REFL judge.DAT likes song.NOM
 ‘which song a/the judge would like.’

4.5.2 Hypothesis and prediction

If objects of object-experiencer predicates are base-generated in SpecvP and their subjects in the complement of V (following the spirit of Belletti & Rizzi 1988 and much subsequent literature), we expect to see the inverse of the subject–object asymmetry witnessed so far (see also the contrast between the prediction profile of H1 and H2c in (14) in section 3). In terms of the statistical model, we expect to see a main effect of CONSTITUENT ORDER,

includes Pesetsky (1995); Rozwadowska (2017); Hirsch (2018); Gogłóza (2021); Antonyuk (2022); Masloch & Poppek & Kiss (2024).

²¹ Case manipulation is inherently tied to the choice the lexical verb, making a within-item manipulation impossible.

²² Experiencers do not have to be humans, they can be animals, too. This would be consistent with manipulating humanness vs. nonhuman animacy. We leave this for future occasions.

²³ We note that accusative-experiencer predicates are less common than dative-experiencer predicates and, moreover, are typically ambivalent between stative predicates and causative predicates, where the subject is construed as a causer rather than theme. Also for this reason, we analyze the dative set and the accusative set separately.

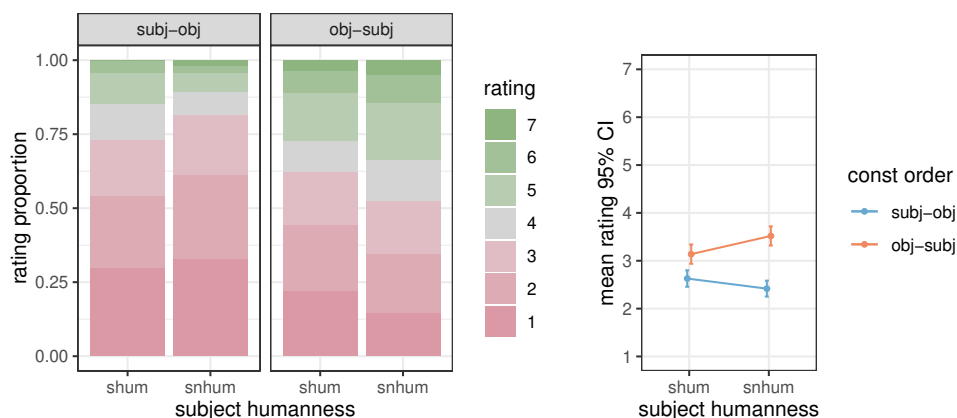


Figure 5: Results of the dative-experiencer part of Experiment 4: proportion of individual ratings in each single condition (left) and mean ratings in each single condition (right)

but in the opposite direction to what we have witnessed so far: extraction from subjects across objects should be more natural than extraction from objects across subject. The manipulation of the factors SUBJECT HUMANNES and CASE is not tied to any particular hypotheses; we have included them for exploratory reasons.

4.5.3 Results and discussion

Results: The results are visualized in Figure 5 (the dative-experiencer part) and Figure 6 (the accusative-experiencer part). Due to the obviously differing results, we have fitted a separate model for each data subset. In both models, we included CONSTITUENT ORDER (treatment-coded; reference level: subj-obj), SUBJECT HUMANNES (sum-coded), and their interaction as fixed effects and random intercepts and slopes for participants and items as random effects. The dative model revealed a main effect of CONSTITUENT ORDER ($z = 4.507, p < .001$): extraction from subjects across objects is more natural than the opposite. In addition, the model showed an interaction between CONSTITUENT ORDER and SUBJECT HUMANNES ($z = -3.797, p < .001$): the effect of CONSTITUENT ORDER is more pronounced if the subject is nonhuman than when it is human. In contrast, the accusative model revealed a main effect of SUBJECT HUMANNES (a slightly more natural extraction from nonhuman subjects than from human subjects; $z = 2.250, p = 0.025$), no effect of CONSTITUENT ORDER ($z = 0.950, p = 0.34$), and no interaction ($z = -1.082, p = 0.28$).

Discussion: The results of the dative-experiencer part of Experiment 4 are consistent with the prediction of the analysis of object-experiencer predicates in which experiencers are base-generated higher (specifically in SpecvP) than subject themes (Belletti & Rizzi 1988). The results show clearly that subextraction from subject themes across object experiencers is more natural (it is, by hypothesis, grammatical) than subextraction from object experiencers across subject themes (which is, by hypothesis, ungrammatical). This is an inverse form of the traditionally observed subject-object asymmetry in subextraction: the first and necessary step of the contemplated subextraction from object experiencers across subjects, the subject would first have to scramble over the object experiencer, to a higher SpecvP, which, however, makes the subextraction impossible – either due to a linearization contradiction or to the ban on spec-to-spec movement, as

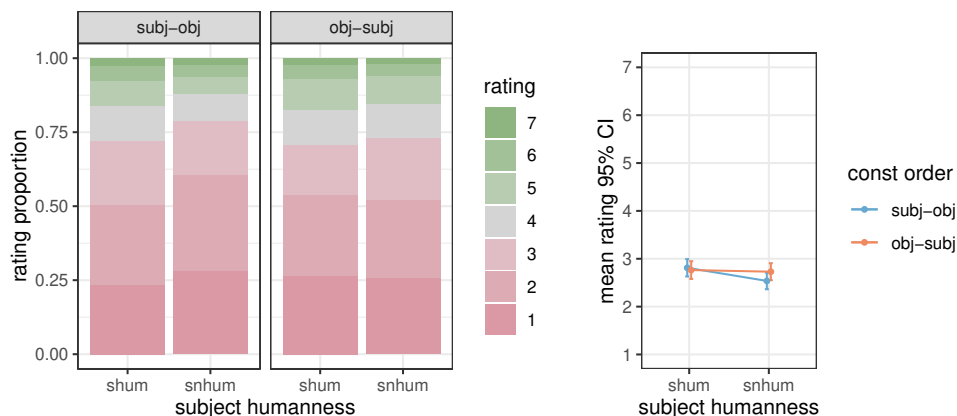


Figure 6: Results of the accusative-experiencer part of Experiment 4: proportion of individual ratings in each single condition (left) and mean ratings in each single condition (right)

detailed for the classical subject–object asymmetry in section 2.1. Besides the main effect of CONSTITUENT ORDER, which has been predicted, we also see an interaction with SUBJECT HUMANNES: the effect of CONSTITUENT ORDER is stronger if the subject is nonhuman than if it is human. In fact, a pairwise post-hoc, performed with *emmeans*, shows that while the contrast between subj-obj and obj-subj is significant within the snhum level ($z = -5.392, p < .001$), it is only marginally significant within the shum level ($z = -2.541, p = 0.054$). It is not clear to us why subextraction in dative-experiencer predicates (with invariably human experiencers) should depend on the humanness of the subject. It is not easy to imagine that ‘liking something’ vs. ‘liking somebody’ would make a difference for argument structure (though see Antonyuk 2022, who, however, couches her analysis in a different set of assumptions, not easily reconcilable with the main finding of Experiment 4). We leave this issue for future research.

Let us now briefly turn to the accusative-experiencer part of Experiment 4. The results of this part of the experiment do not support our hypothesis: we see no subject–object asymmetry in either direction. The effect of SUBJECT HUMANNES – a slightly more natural extraction from a nonhuman subject than from a human one (possibly reminiscent of what we have seen for dative experiencers, too) is probably spurious, as a post-hoc pairwise test has not revealed any significant contrast whatsoever (all $ps > .1$). We suspect that the results for the accusative-experiencer predicates are unclear, esp. when compared to the results in the dative-experiencer part, because most of the subjects are ambiguous between themes and causers or agents (if human). If the subject is interpreted as a causer or agent, the object is construed as a patient, which in turn affects the argument structure and subextraction (un)availability. The ambiguity (and hence processing difficulty) not only pushes the naturalness down (none of the conditions is rated with a mean higher than 3), but also blurs any potential interaction with subextraction. A more nuanced design would be needed to disentangle the ambiguity.

4.6 Summary and general discussion

After reporting experimental evidence for subject–object asymmetry in left branch extraction in Czech (due to the hitherto unpublished work of Šimík and colleagues), noticed previously from Japanese (e.g. Saito 1985), Korean (e.g. Ko 2007), or Russian (Bon-

darenko & Davis 2021), we set out to address two research questions concerning the relationship between humanness and syntax: Are humanness related word order alternations represented in narrow syntax, or are they mere surface preferences (as suggested, e.g., by Fanselow 2012)? If they are part of narrow syntax, what syntactic process do humanness-related alternations correspond to?

In a series of experiments, we have argued that humanness-related word order manipulations do indeed stem from narrow syntax, corroborating syntactic approaches to humanness (or, more generally, animacy) based mostly on morphological evidence (such as agreement, Woolford 2000; Deal & Royer *online first*, or differential object marking, Fábregas 2013, among many others). Moreover, relying on the phenomenon of subject-object asymmetry in subextraction, we have formulated specific predictions of three syntactic analyses representing different views of how exactly humanness affects argument structure.

Our main experiment – Experiment 1 (section 4.2) – provided evidence that human objects obligatorily scramble over nonhuman subjects, specifically to a higher SpecvP. The consequence of this scrambling is the impossibility to perform subextraction both from subjects across objects (the standard case of the aforementioned subject-object asymmetry), but also from human objects across nonhuman subjects. Experiment 2 (section 4.3) showed that when humanness of both arguments is balanced, that is, when both the subject and object are either human, or nonhuman, the standard subject-object asymmetry applies. Experiment 3 (section 4.4) showed that the ban on extraction from a human object across a nonhuman subject is lifted when the nonhuman subject is determinate (more specifically, anaphoric demonstrative NP). This follows from hierarchical multifactorial accounts of word order phenomena (Müller 1999; Titov 2012), where some factors are stronger than others. In the case at hand, determinacy is a stronger factor than humanness, which means that an indeterminate human object does not scramble over a determinate nonhuman subject, effectively making extraction from the object possible. It remains to be seen whether other factors, specifically information structure, are even stronger than determinacy (see esp. Titov 2012 for discussion). One specific prediction to explore, for instance, is that extraction from a human determinate object across nonhuman indeterminate, but topical or discourse-given subject should be possible. This is left for future research. Our final experiment – Experiment 4 (section 4.5) explored a further prediction, namely that the subject-object asymmetry reverses with the use of object-experiencer predicates. One part of the experiment, using dative experiencers (vs. accusative ones) delivered corroborating and promising results: we do indeed see that extraction from subjects across object experiencers (which are typically human) is more natural than extraction from object experiencers across subjects.

A striking phenomenon present in all of our experiments was the general low naturalness of the stimuli. The conditions which are – by assumption – grammatical have generally reached a median rating of 3 on a 7-point Likert scale (with 7 most natural). We have assumed that the low rating is due to the overall complexity of the structures under scrutiny and consequently difficult processing. If correct, our experiments are a good example of the impossibility to interpret Likert scale-based acceptability (or naturalness) in an absolute fashion: there is nothing like a universally valid “acceptable” vs. “unacceptable” area. Grammaticality differences can correspond to relatively small absolute rating differences and can be manifested within a relatively “unacceptable area” of the scale. In addition, acceptability or naturalness are necessarily very coarse measures, reflecting effects from all areas of grammar, semantics, pragmatics, and processing.

This makes carefully designed experiments essential for reaching a better and complex understanding of grammatical phenomena.

5 Conclusion and open issues

Humanness is a semantic property of nouns or nominal phrases which has been observed to interact in non-trivial ways with morphology and syntax (Woolford 2000; Fábregas 2013; Deal & Royer *online first*; a.o.). The effect of humanness (or, more generally, animacy) on word order has also been relatively well-documented (Müller 1999; Fanselow 2012; Titov 2012; Antonyuk 2022; Seržant et al. *in press*; a.o.), but its grammatical and narrowly syntactic status has been controversial. This is in part due to the fact that word order itself is affected by a multitude of factors (see Jasinskaja & Šimík *forthcoming* for a survey about Slavic languages), whose effects appear to be in the realm of tendencies rather than categorical rules and whose application seems optional rather than obligatory. That in turn invites surface-oriented and processing-related accounts.

The present paper constitutes a complex argument from Czech, backed by controlled naturalness rating experiments, that the effects of humanness on word order are of narrowly syntactic origins. More specifically, we argued that human objects must scramble across nonhuman subjects, targeting a specifier of vP (unless the scrambling is overridden by a stronger requirement, such as the subject's determinacy, which we saw evidence for in Experiment 3). The obligatory scrambling, along with independent analytical assumptions about phase-based syntax (Chomsky 2001; Fox & Pesetsky 2005; Ko 2007), correctly predicts the impossibility of left branch extraction from human objects across nonhuman subjects, which would be permitted in the absence of the scrambling.

The present paper offered a detailed case study of the impact of humanness on word order. It also provided the first (to the best of our knowledge) experimental evidence for the subject-object asymmetry in subextraction (previously noticed for Japanese, Korean, or Russian; Saito 1985; Ko 2007; Bondarenko & Davis 2021) and for a very specific conjunction of assumptions about how the syntax of natural language operates. At the same time, it has suggested new directions of empirical research in the area of word order, esp. with regard to other factors that have been argued to influence word order, such as determinacy or information structure, and, equally importantly, their mutual dependencies and/or interactions. As we already suggested above, we would expect, for instance, that extraction from determinate human objects across indeterminate nonhuman subjects should be possible, as long as the subject is topical or discourse-given. What has been left untouched in the present study is left branch extraction across other types of arguments or adjuncts. In some cases, the predictions seem relatively clear and uncontroversial. If, for instance, two internal arguments originate within the VP, as is usually assumed, left branch extraction from one object across another should be unproblematic. On the other hand, the presently entertained account predicts extraction to be impossible from an external argument across a manner or agent-related adjuncts. These and other non-trivial predictions can be tested in future work and help further understanding of syntax, word order, and the factors that affect them.

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